

EXPLORATORY DATA ANALYSIS ON E-COMMERCE SALES DATASET

Project Report

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Disclosure

This project report titled “Exploratory Data Analysis on E-Commerce Sales Dataset” has been prepared by me during my Data Analytics Internship at DecodeLabs.

All content included in this report is original and has not been plagiarized by me except for reference purposes only.

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Date: _____2026/05/25_____

Acknowledgement

I am grateful to DecodeLabs for granting me the chance to do this Exploratory Data Analysis project in the context of the Data Analytics Internship program.

I wish to thank the mentors and instructors for their valuable guidance and help during this project.

In this project, I got a chance to enhance my practical skills in data analysis and visualization in Python.

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1. Introduction

This work revolves around performing exploratory data analysis (EDA) on an e-commerce sales dataset using Python and visualization libraries in Google Collab.

Exploratory data analysis (EDA) is the practice of identifying insights within raw data. Various analytical procedures were used to gain insights into customer behavior, sales performance, payment methods, and product trends.

Data cleaning procedures, such as dealing with missing data, were also performed before performing the analysis.

2. Objectives

- To understand the structure of the dataset
- To identify missing values and duplicate records
- To perform data cleaning and preprocessing
- To analyze sales and customer behavior
- To visualize important trends and patterns
- To generate insights from the dataset

3. Technologies Used

- Python - Data analysis
- Pandas - Data manipulation
- Matplotlib - Data visualization
- Seaborn - Statistical visualization
- Google Colab - Development environment

4. Dataset Description

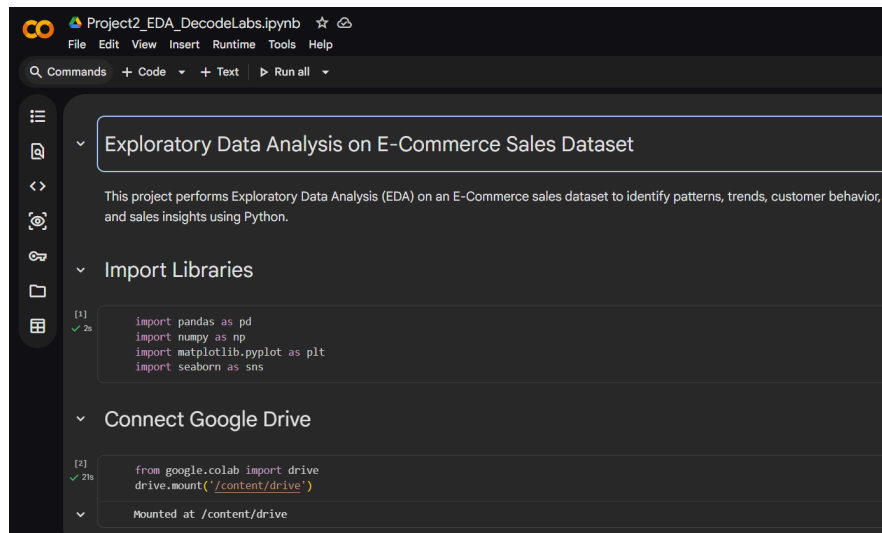
E-commerce sales transaction data was used for this project. The dataset contains information about products, payments, orders, transactions, and activities related to sales.

The dataset was examined using data mining techniques to find patterns between different data items.

5. Screenshot

Figure 1: Importing Required Libraries

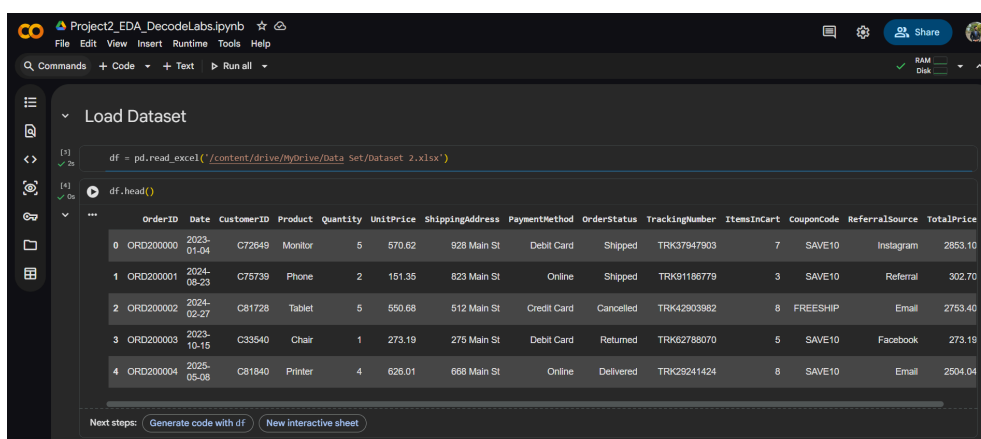
This figure shows the libraries imported for data analysis and visualization. Pandas was used for data manipulation, while Matplotlib and Seaborn were used to create charts and visual representations of the dataset.



```
Project2_EDA_DecodeLabs.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Exploratory Data Analysis on E-Commerce Sales Dataset
This project performs Exploratory Data Analysis (EDA) on an E-Commerce sales dataset to identify patterns, trends, customer behavior, and sales insights using Python.
Import Libraries
[1] ✓ 2s
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
Connect Google Drive
[2] ✓ 21s
from google.colab import drive
drive.mount('/content/drive')
Mounted at /content/drive
```

Figure 2: Loading the Dataset / Dataset Inspection / Dataset Information

- This figure shows the process of loading the E-Commerce sales dataset into Google Colab for analysis and preprocessing tasks.
- This figure displays the first few rows of the dataset. It helped to understand the dataset structure, available columns, and the type of information stored in each record.
- This figure shows the dataset information, including the number of rows, columns, data types, and non-null values. It was useful for identifying missing values and understanding the dataset structure before analysis.



```
Project2_EDA_DecodeLabs.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Load Dataset
[3] ✓ 2s
df = pd.read_excel('/content/drive/MyDrive/Data Set/Dataset 2.xlsx')
[4] ✓ 0s
df.head()
OrderID Date CustomerID Product Quantity UnitPrice ShippingAddress PaymentMethod OrderStatus TrackingNumber ItemsInCart CouponCode ReferralSource TotalPrice
0 ORD200000 2023-01-04 C72649 Monitor 5 570.62 928 Main St Debit Card Shipped TRK37947903 7 SAVE10 Instagram 2853.10
1 ORD200001 2024-08-23 C75739 Phone 2 151.35 823 Main St Online Shipped TRK91186779 3 SAVE10 Referral 302.70
2 ORD200002 2024-02-27 C81728 Tablet 5 550.68 512 Main St Credit Card Cancelled TRK42903882 8 FREESHIP Email 2753.40
3 ORD200003 2023-10-15 C33540 Chair 1 273.19 275 Main St Debit Card Returned TRK62788070 5 SAVE10 Facebook 273.19
4 ORD200004 2025-05-08 C81840 Printer 4 626.01 668 Main St Online Delivered TRK29241424 8 SAVE10 Email 2504.04
Next steps: Generate code with df New Interactive sheet
```

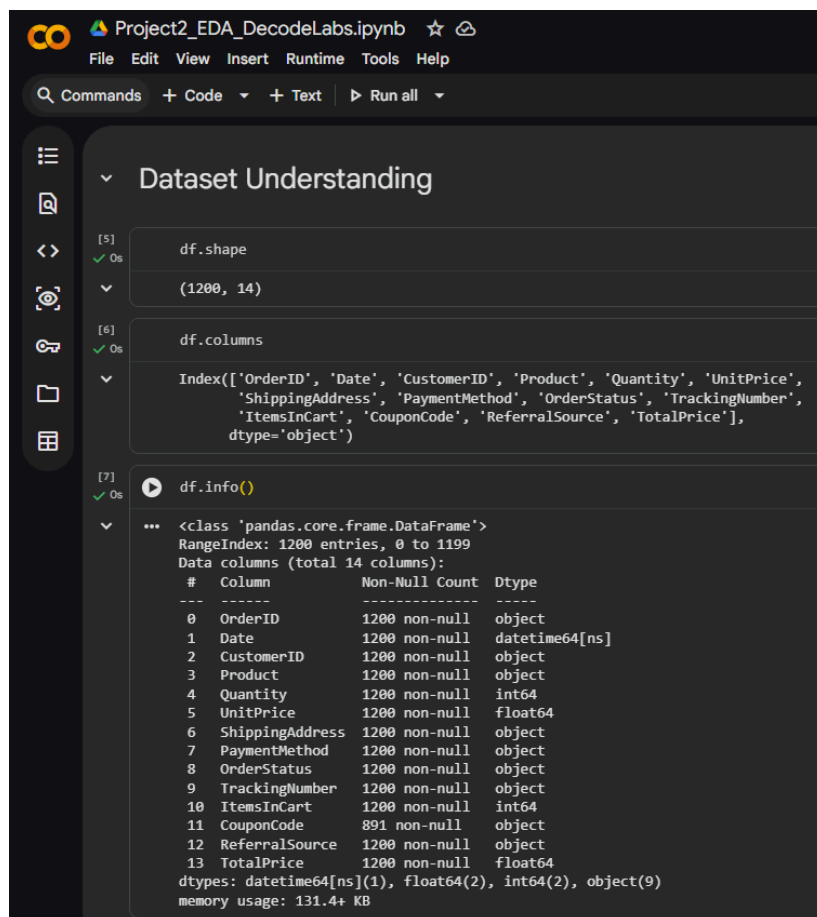


Figure 3: Missing Value Detection

This figure shows the missing values identified in the dataset. Detecting missing values is an important step in data preprocessing because incomplete data can affect the accuracy of analysis results.

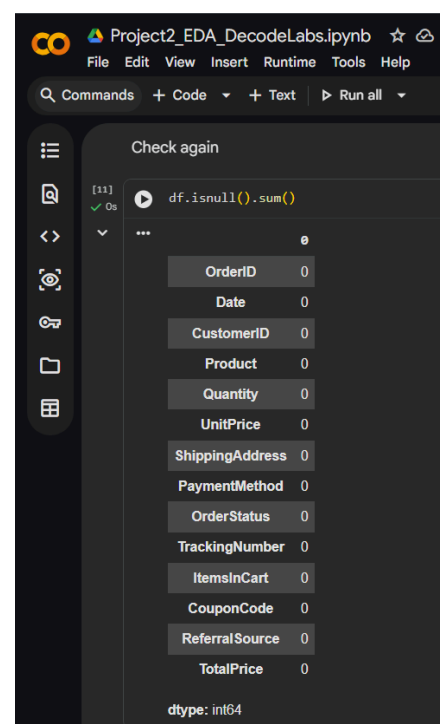
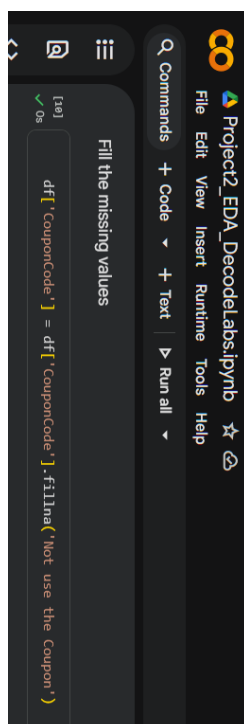
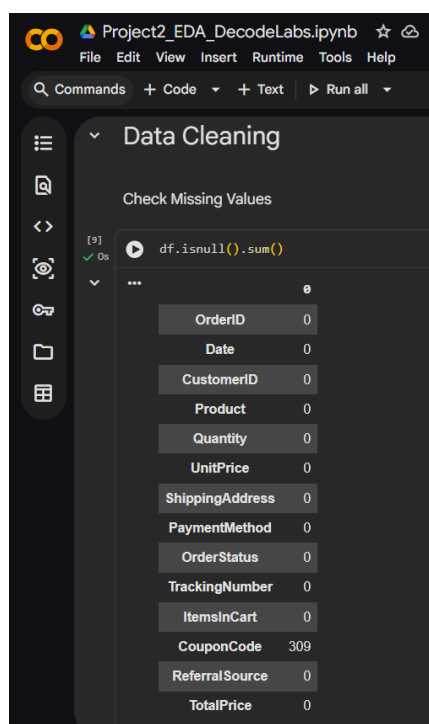
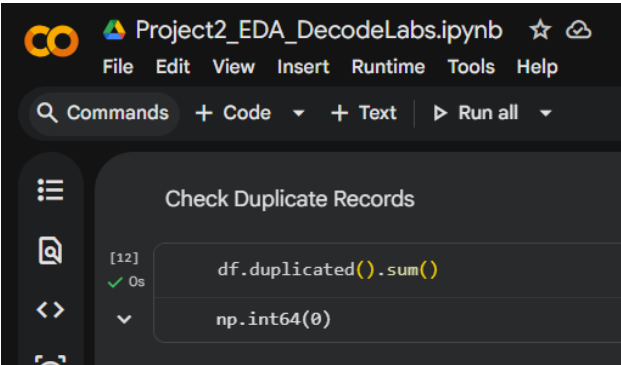


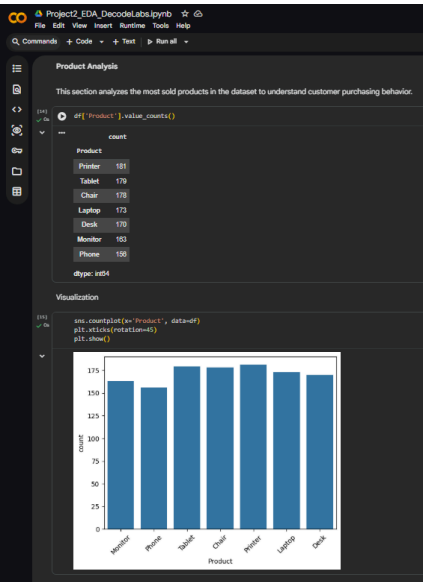
Figure 4: Duplicate Record Checking

This figure shows the process of checking duplicate records in the dataset. Duplicate records were analyzed to maintain data consistency and avoid incorrect analytical results.



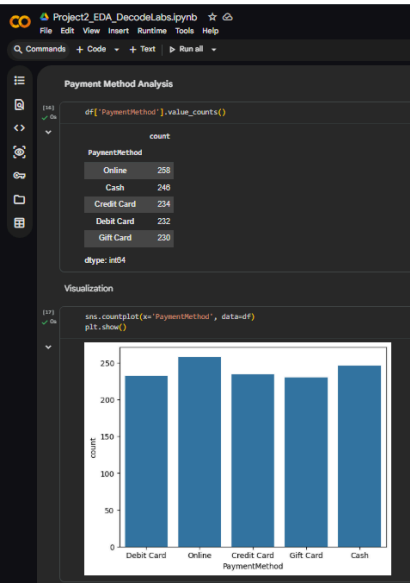
Analysis Process

Figure 5: Product Analysis



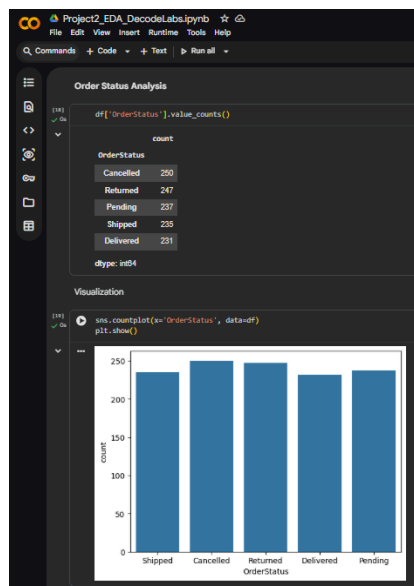
This figure represents the analysis of product-related sales data. The visualization helped identify which products were purchased more frequently and showed differences in product performance.

Figure 6: Payment Method Analysis



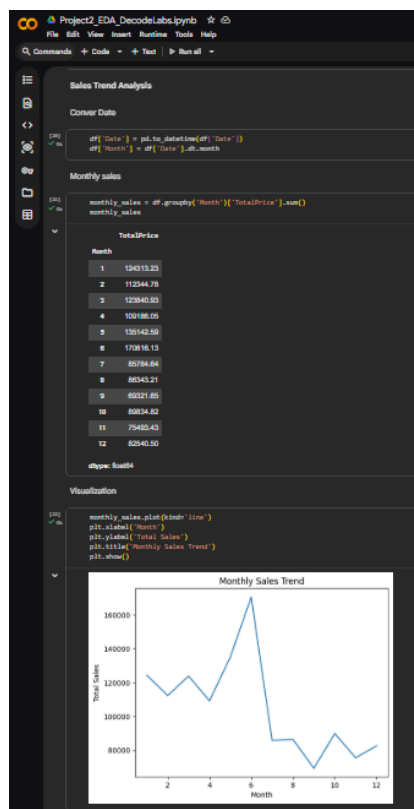
This figure shows the distribution of payment methods used by customers. The analysis provided insights into the most commonly used payment options in customer transactions.

Figure 7: Order Status Analysis



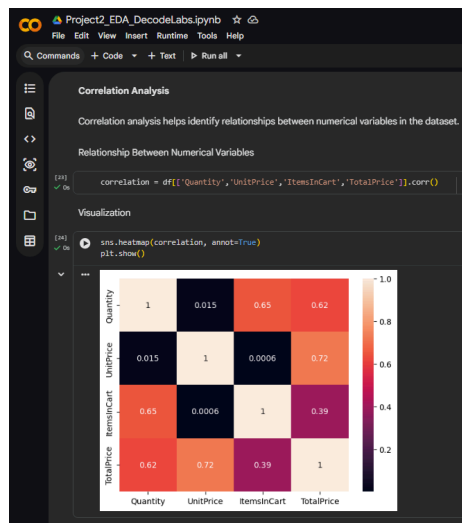
This figure illustrates the different order statuses available in the dataset. It helped analyze completed, pending, or canceled orders and understand overall order processing trends.

Figure 8: Monthly Sales Trend Analysis



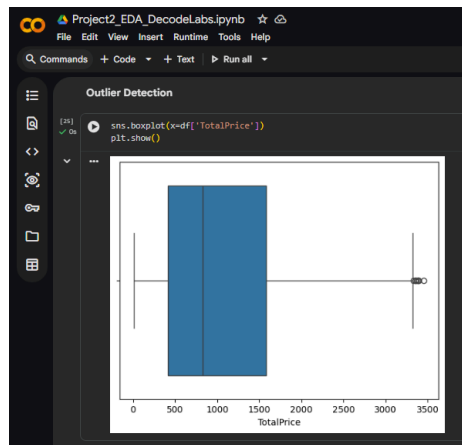
This figure presents the monthly sales trend observed in the dataset. The line chart helped identify fluctuations and patterns in sales performance over time.

Figure 9: Correlation Heatmap



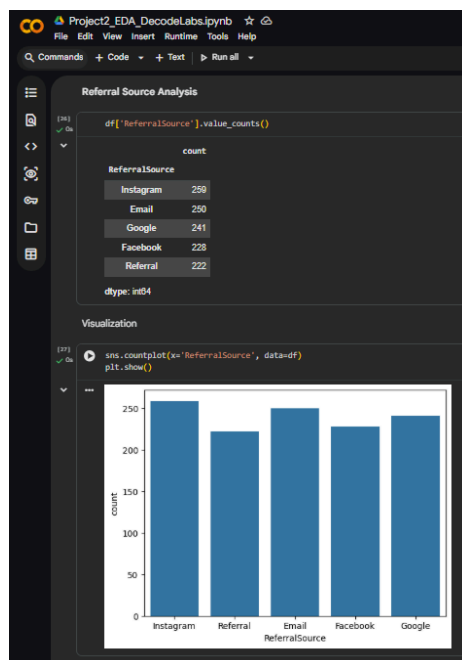
This figure displays the correlation heatmap created for numerical variables in the dataset. The heatmap helped identify relationships and dependencies between different variables.

Figure 10: Outlier Detection



This figure shows the boxplot used for outlier detection. Outliers were analyzed to identify unusual values that may affect the accuracy of data analysis and visualization results.

Figure 11: Referral Source Analysis



This figure represents the analysis of referral sources in the dataset. The visualization helped identify the platforms or channels contributing to customer visits and sales activity.